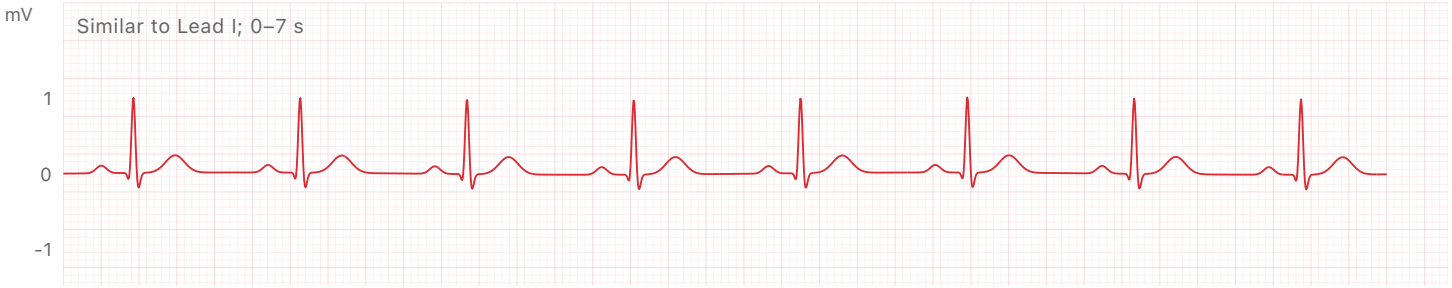
Record export

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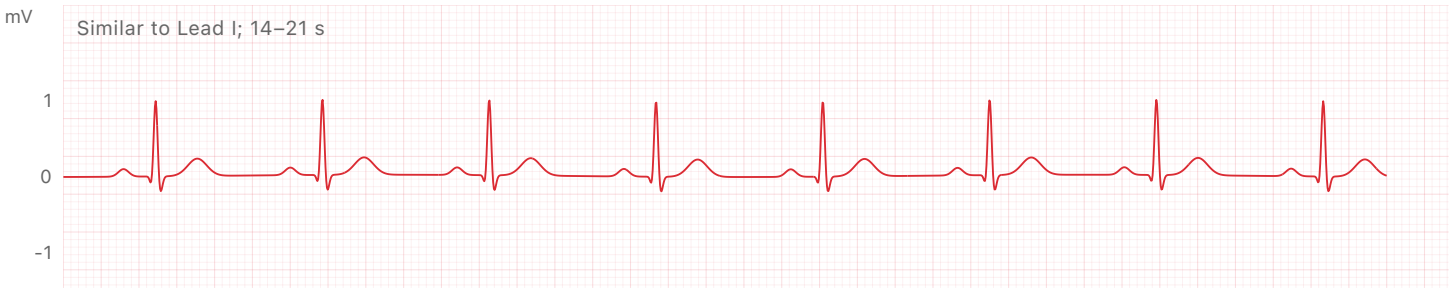
Apple Watch

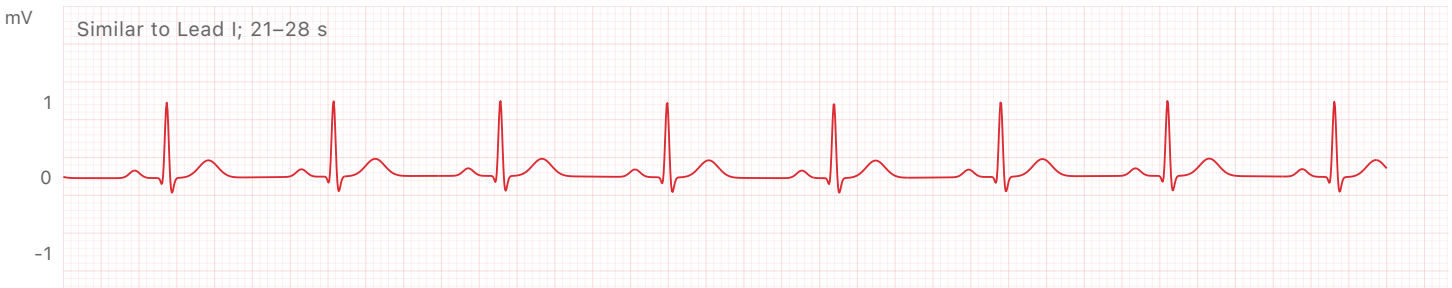
Heart rate	Recording duration	Demo classification
68 bpm	30 s	Sinus rhythm

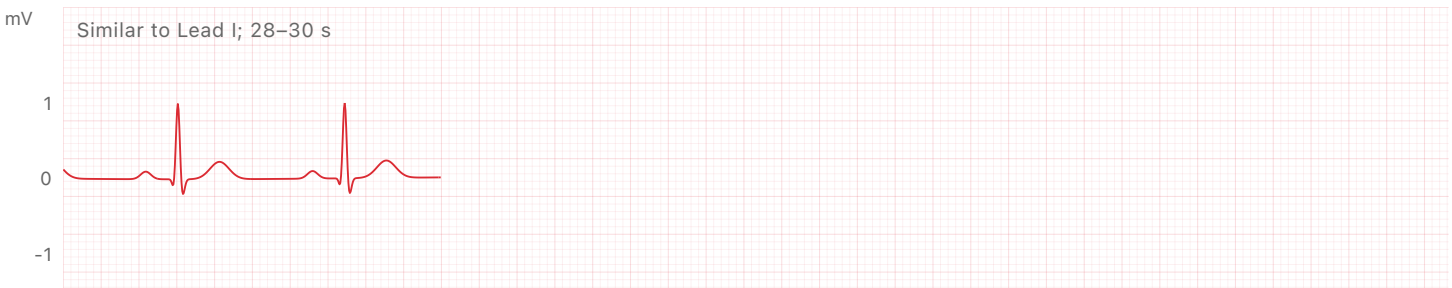
25 mm/s 10 mm/mV Print at actual size (100%)











For reviewing records only. Not medical advice or a substitute for professional medical care.

Analysis summary

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Average heart rate in this record	Signal coverage	Analyzed duration
68 bpm	100%	30 s

Metric	Result
Data sources	Apple Watch / Synthetic example
R peaks identified	34
Median RR	883 ms
RR standard deviation	1 ms
RMS of successive RR differences	2 ms

RR statistics do not exclude non-sinus beats and are not normal-beat HRV measures.

HRV from this record

SDNN	RMSSD	SD1 / SD2
1 ms	2 ms	1 / 1 ms

27 seconds of clear signal, 31 beat intervals

This HRV estimate is from a short recording. Compare recordings of the same length taken under similar conditions.

Interval statistics can be calculated from short recordings, but reliability depends on the metric, recording length and conditions. Study findings differ; this estimate is not equivalent to standardized five-minute HRV. [H1] [H2] [H3]

Premature-beat candidates and rhythm

Rhythm reference score	Premature-beat candidates
100 / 100	Beats: 0

Detected beats: 34; included in analysis: 32.
Flagged beats: 0. Each beat is counted once.

Automatic flags require review against the original ECG. The rhythm score does not assess heart health. [R1] [E1]

Segment measurements

Range is the minimum to maximum in this recording, not a clinical reference interval. The central 80% is P10–P90.
— Unable to measure

Segment measurements

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P wave

Measurement	Median	Range	P10–P90	Valid / total
P-wave duration	121 ms	117–121 ms	117–121 ms	34 / 34
P-wave amplitude	0.10 mV	0.10–0.10 mV	0.10–0.10 mV	34 / 34

[E2] [E1]

PR interval

Measurement	Median	Range	P10–P90	Valid / total
PR interval	188 ms	188–191 ms	188–188 ms	34 / 34

[E2] [E1]

PR segment

Measurement	Median	Range	P10–P90	Valid / total
PR-segment duration	66 ms	66–70 ms	66–70 ms	34 / 34
PR-segment level	0.00 mV	0.00–0.00 mV	0.00–0.00 mV	34 / 34

[E2] [E1]

QRS complex

Measurement	Median	Range	P10–P90	Valid / total
QRS duration	98 ms	94–98 ms	94–98 ms	34 / 34
Q amplitude (estimated)	-0.07 mV	-0.07–-0.06 mV	-0.07–-0.06 mV	34 / 34
Q duration (estimated)	20 ms	16–20 ms	17–20 ms	34 / 34
Main R-peak amplitude	0.94 mV	0.92–0.94 mV	0.93–0.94 mV	34 / 34
QRS start to R peak	43 ms	39–43 ms	39–43 ms	34 / 34
S amplitude (estimated)	-0.18 mV	-0.18–-0.18 mV	-0.18–-0.18 mV	34 / 34
S duration (estimated)	31 ms	31–35 ms	31–35 ms	34 / 34

[E1]

Segment measurements

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ST segment / J point

Measurement	Median	Range	P10–P90	Valid / total
ST-segment duration	59 ms	55–59 ms	55–59 ms	34 / 34
J-point level	-0.01 mV	-0.01–-0.00 mV	-0.01–-0.00 mV	34 / 34
Level 60 ms after J	—	—	—	0 / 34
Level 80 ms after J	—	—	—	0 / 34
ST voltage change rate	—	—	—	0 / 34

Reasons for missing values: Occurrences: 34. The T-wave start is unclear, so the ST measurement window cannot be determined.; Occurrences: 34. This position is too close to the T wave to measure ST separately.

[E1] [E3]

T wave

Measurement	Median	Range	P10–P90	Valid / total
T-wave duration	219 ms	219–223 ms	219–219 ms	34 / 34
T-wave amplitude	0.23 mV	0.23–0.23 mV	0.23–0.23 mV	34 / 34
T-wave duration ratio	1.00 ×	0.97–1.00 ×	1.00–1.00 ×	34 / 34

[E1] [E3]

QT interval

Measurement	Median	Range	P10–P90	Valid / total
QT interval	371 ms	371–375 ms	371–375 ms	34 / 34

[E1] [E3]

QTc (corrected QT)

Measurement	Median	Range	P10–P90	Valid / total
QTc (Fridericia)	387 ms	387–391 ms	387–391 ms	31 / 34
QTc (Bazett)	395 ms	395–399 ms	395–399 ms	31 / 34

Reasons for missing values: Occurrences: 2. The QT end is unclear, or nearby RR intervals do not meet the correction conditions. See Measurement method for details.; Occurrences: 1. This beat's QT or nearby continuous, stable RR intervals are missing, so correction is unavailable.

[E1] [E3]

Segment measurements

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TP segment

Measurement	Median	Range	P10–P90	Valid / total
TP-segment duration	324 ms	320–324 ms	320–324 ms	33 / 34

Reasons for missing values: Occurrences: 1. This T-wave end or the next P-wave start is unclear, or there may be a signal gap between them.

[E1]

RR / PP intervals

Measurement	Median	Range	P10–P90	Valid / total
RR interval	883 ms	879–883 ms	880–883 ms	33 / 34
PP interval	883 ms	879–883 ms	880–883 ms	33 / 34

Reasons for missing values: Occurrences: 1. The previous beat is missing, the P wave is unclear, or the signal has a gap between beats.

[E1] [R2]

Measurement method

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These values are estimated automatically by the app. Accuracy has not been validated on real Apple Watch data, and the values cannot determine whether a record is normal.

Detection thresholds, quality filters, candidate labels and the rhythm reference score are HeartPro's own rules, not medical diagnostic standards.

Metric	Result
Data sources	Apple Watch
Sampling rate	256 Hz
Analysis version	0.6.0-research

Detection and scoring rules

Rhythm reference score = 100 × (1 – flagged beats ÷ reviewable beats), rounded. Flags include premature-beat candidates, marked interval changes or local waveform changes. The pause after a candidate is not counted again. This describes the share of beats not flagged by these rules in this recording. It is not a heart-health score, and 100 does not mean the ECG is normal.

At least 20 seconds of clear signal, 80% signal coverage, 15 reviewable beats and 8 stable reference beats are required. At least 80% of non-edge beats must be reviewable. Poor signal does not reduce the score; missing PR or QT values and higher or lower HRV do not add or subtract points.

Compare with nearby beats: the beat must arrive over 15% and at least 80 ms early, with a longer following interval or a wider, changed QRS. Wide candidates require a local width of at least 120 ms, over 20% wider than the reference, and template correlation below 0.90. Narrow candidates require a width under 120 ms, correlation of at least 0.90 and a longer following interval. These research rules cannot by themselves confirm atrial or ventricular premature beats.

Rule version: rhythm-review-1

P wave

The start, end and peak of the P wave are estimated. If it is too small, its boundaries are unclear or it overlaps nearby waves, the affected values are left blank.

PR interval

Uses the P-wave start and QRS start from the same beat. If either is unclear, no PR interval is calculated. PR includes the P wave itself and the following PR segment.

PR segment

Measured from the P-wave end to QRS start. The level is the mean voltage of this segment relative to the local TP baseline. Positive values are above the baseline; negative values are below it.

QRS complex

QRS start and end are estimated first. Opposite-direction waves before and after the main peak are candidates for Q and S. Values retain their original polarity. If the main peak points down, Q, R and S labels need manual review. A single lead cannot assess the electrical axis or R-wave progression across chest leads.

Measurement method

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ST segment / J point

The QRS end estimates the J point. Voltage is measured 60 and 80 ms later. The change rate comes from the voltage difference between these positions: positive means upward and negative downward. Positions too close to the T wave are not used for ST measurements.

T wave

The T-wave start, peak and end are found separately. The duration ratio is start-to-peak time divided by peak-to-end time. It is near 1 when the two parts have similar lengths. Multiple peaks require checking the original trace.

QT interval

Each beat is measured separately, from its QRS start to its T-wave end, before results are summarized. If the T-wave end is unclear, QT is not calculated.

QTc (corrected QT)

Uses QT from the same beat and the immediately preceding RR, both in seconds: Fridericia = QT / RR^(1/3); Bazett = QT / √RR. The algorithm requires continuous neighboring RR intervals, changes no greater than 10%, and intervals of 500–1500 ms. These are algorithm conditions, not a normal range.

TP segment

Measured from this T-wave end to the next P-wave start. No value is produced if either boundary is unclear, the signal has a gap, or the interval is shorter than 40 ms.

RR / PP intervals

RR and PP are calculated from their own consecutive peaks. They do not substitute for one another, and intervals are not joined across signal gaps.

Recording notes

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Duration: 2 min
Caffeine

USERNOTE café 日本語 한글

Health information and sources

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HeartPro is an informational measurement tool for reviewing existing records, not intended for use as a medical device. Automated estimates can be wrong and must not be used for diagnosis, treatment or assessing personal health.

These sources explain health concepts, data sources and calculation methods. They do not validate HeartPro's algorithms or imply endorsement by Apple or the authors.

Detection thresholds, quality filters, candidate labels and the rhythm reference score are HeartPro's own rules, not medical diagnostic standards.

Original waveforms and source classification: [A1] [A2]. System SDNN: [A3]. Wave definitions and methods: [E1] [E2] [E3]. Short-recording HRV: [H1] [H2] [H3]. Heartbeat and heart-rate concepts: [R1] [R2].

[A1] Apple: ECG use and limitations

Apple Support. Take an ECG with the ECG app on Apple Watch.

https://support.apple.com/en-us/120278

[A2] Apple: HealthKit ECG data

Apple Developer. HKElectrocardiogram.

https://developer.apple.com/documentation/healthkit/hkelectrocardiogram

[A3] Apple: SDNN in Health

Apple Developer. heartRateVariabilitySDNN.

https://developer.apple.com/documentation/healthkit/hkquantitytypeidentifier/heartratevariabilitysdnn

[E1] MSD: waveform and interval definitions

Cascino T, Shea MJ. Electrocardiography. MSD Manual Professional Edition. 2026.

https://www.msdmanuals.com/professional/cardiovascular-disorders/cardiovascular-tests-and-procedures/electrocardiography

[E2] StatPearls: P wave and PR interval

Douedi S, Douedi H. P wave. StatPearls. Updated 2023.

https://www.ncbi.nlm.nih.gov/books/NBK551635/

[E3] AHA / ACCF / HRS: ST, T, U and QT

Rautaharju PM et al. ECG standardization, part IV. Circulation. 2009;119:e241–e250. doi:10.1161/CIRCULATIONAHA.108.191096.

https://pubmed.ncbi.nlm.nih.gov/19228821/

[H1] HRV metrics, Poincaré plots and measurement context

Shaffer F, Ginsberg JP. An Overview of Heart Rate Variability Metrics and Norms. Front Public Health. 2017;5:258. doi:10.3389/fpubh.2017.00258.

https://pubmed.ncbi.nlm.nih.gov/29034226/

Health information and sources

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[H2] Ultra-short HRV: 2015 study

Munoz ML et al. Validity of (Ultra-)Short Recordings for Heart Rate Variability Measurements. PLoS ONE. 2015;10:e0138921. doi:10.1371/journal.pone.0138921.

https://pubmed.ncbi.nlm.nih.gov/26414314/

[H3] Ultra-short HRV: 2021 reliability study

Burma JS et al. The validity and reliability of ultra-short-term heart rate variability parameters and the influence of physiological covariates. J Appl Physiol. 2021;130:1848–1867. doi:10.1152/japplphysiol.00955.2020.

https://pubmed.ncbi.nlm.nih.gov/33856258/

[R1] AHA: premature heartbeats

American Heart Association. Premature Contractions — PACs and PVCs. 2024.

https://www.heart.org/en/health-topics/arrhythmia/about-arrhythmia/premature-contractions-pacs-and-pvcs

[R2] AHA: heart rate and influencing factors

American Heart Association. All About Heart Rate (Pulse). 2024.

https://www.heart.org/en/health-topics/high-blood-pressure/the-facts-about-high-blood-pressure/all-about-heart-rate-pulse

Data protection

HeartPro does not collect your health records, notes or analysis results. They are processed and stored only on your device and are not automatically uploaded to developer or third-party servers. There are no accounts, ads, tracking or third-party analytics SDKs.

Only your explicit actions share a report with your chosen destination through the system share sheet or save selected symptoms to Apple Health. HeartPro does not sync health data through iCloud; Apple Health's own synchronization is controlled by system settings.